

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12415911
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Perdomi; Gianni et al.

Polyethylene composition for filaments or fibers

Abstract

A polyethylene composition made from or containing: A) from 75% to 97% by weight of a copolymer of ethylene having: 1) a density of 0.925 g/cm.³ or higher; and 2) a MI._{sub.2} value of 0.5 g/10 min. or greater; and B) from 3% to 25% by weight of a polyolefin composition made from or containing: B._{sup.I}) from 5% to 35% by weight of a propylene homopolymer or a propylene copolymer; B._{sup.II}) from 20% to 50% by weight of a copolymer of ethylene containing from 1.0% to 20.0% by weight of alpha-olefin units and 25.0% by weight or less of a fraction soluble in xylene at 25° C.; and B._{sup.III}) from 30% to 60% by weight of a copolymer of ethylene and propylene containing from 25.0% to 75.0% by weight of ethylene derived units and from 40.0% to 95.0% by weight of a fraction soluble in xylene at 25° C.

Inventors:	Perdomi; Gianni (Ferrara, IT), Galvan; Monica (Ferrara, IT)
Applicant:	BASELL POLYOLEFINE GMBH (Wesseling, DE)
Family ID:	1000008818415
Assignee:	Basell Polyolefine GmbH (Wesseling, DE)
Appl. No.:	17/779773
Filed (or PCT Filed):	December 02, 2020
PCT No.:	PCT/EP2020/084319
PCT Pub. No.:	WO2021/110769
PCT Pub. Date:	June 10, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230075830 A1	Mar. 09, 2023

Foreign Application Priority Data

Publication Classification

Int. Cl.: C08L23/0807 (20250101); D01F6/26 (20060101); D01F6/30 (20060101)

U.S. Cl.:

CPC C08L23/0815 (20130101); D01F6/30 (20130101); C08L2203/12 (20130101); C08L2205/025 (20130101); C08L2205/035 (20130101); D10B2321/021 (20130101); D10B2505/202 (20130101)

Field of Classification Search

USPC: None

References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4399054	12/1982	Ferraris et al.	N/A	N/A
2013/0064995	12/2012	Perdomi et al.	N/A	N/A
2016/0137824	12/2015	Kohler et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
45977	12/1981	EP	N/A
0129368	12/1983	EP	N/A
0361493	12/1989	EP	N/A
0416815	12/1990	EP	N/A
0545304	12/1992	EP	N/A
0561479	12/1992	EP	N/A
0576970	12/1993	EP	N/A
0632063	12/1994	EP	N/A
0659758	12/1994	EP	N/A
0661300	12/1994	EP	N/A
728769	12/1995	EP	N/A
0031090	12/1999	WO	N/A
2005005730	12/2004	WO	N/A
2009101124	12/2008	WO	N/A
2009103516	12/2008	WO	N/A
2011134897	12/2010	WO	N/A
WO-2017202600	12/2016	WO	C08L 23/0815
2019007684	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion Mailed Feb. 26, 2021 (Feb. 26, 2021) For Corresponding PCT/EP2020/084319. cited by applicant

Primary Examiner: Rioja; Melissa A

Assistant Examiner: Darling; Devin Mitchell

Background/Summary

(1) This application is the U.S. National Phase of PCT International Application PCT/EP2020/084319, filed Dec. 2, 2020, claiming benefit of priority to European Patent Application No. 19213059.9, filed Dec. 3, 2019, the contents of which are incorporated herein by reference in its entirety.

FIELD OF THE INVENTION

(2) In general, the present disclosure relates to the field of chemistry. More specifically, the present disclosure relates to polymer chemistry. In particular, the present disclosure relates to a polyethylene composition for filaments or fibers.

BACKGROUND OF THE INVENTION

(3) As used herein, the term “filaments” refers to fibers for textile and carpeting applications. As used herein, the term “artificial turf filaments” refers to filaments having a titer of at least 20 denier (hereinafter called “den”) and used to prepare artificial turf structures.

(4) In some instances and to produce the final artificial turf structure, the filaments are fixed to a backing substrate.

(5) In some instances, softness and UV resistance of artificial turf filaments are achieved by using polyethylene materials as a polyolefin component of the filament or fiber.

(6) In some instances, the polyethylene materials are produced by using metallocene catalysts in polymerization.

(7) In some instances, polymer filaments or fibers are made from or containing ethylene polymers having: a) a density of 0.900 g/cm.^{sup.3} or higher; b) a MI._{sub.21} value of 25 g/10 min. or greater; and c) a MFR value from 25 to 60.

(8) In some instances, the ethylene polymers are mixed with an elastomeric or plastomeric polyolefin or polyolefin composition.

SUMMARY OF THE INVENTION

(9) In a general embodiment, the present disclosure provides a polyethylene composition, hereinafter called “polyethylene composition (I)”, made from or containing: A) from 75% to 97% by weight of a copolymer of ethylene with one or more comonomers selected from C._{sub.3}-C._{sub.10} alpha-olefins, wherein the ethylene copolymer has: 1) a density of 0.925 g/cm.^{sup.3} or higher measured according to ISO 1183-1:2012 at 23° C.; and 2) a MI._{sub.2} value of 0.5 g/10 min. or greater, where MI._{sub.2} is the Melt Index measured according to ISO 1133-2:2011 at 190° C. with a load of 2.16 kg; and B) from 3% to 25% by weight of a polyolefin composition made from or containing: B._{sup.I}) from 5% to 35% by weight of one or more propylene homopolymer(s) or one or more copolymer(s) of propylene with ethylene and/or C._{sub.4}-C._{sub.8} α-olefin(s), containing 90% or more of propylene units, or combinations of the homopolymers and copolymers; and component B._{sup.I}) contains 10.0% by weight or less of a fraction soluble in xylene at 25° C. (XS._{sup.I}), both the amount of propylene units and of the fraction XS._{sup.I} being referred to the weight of B._{sup.I}); B._{sup.II}) from 20% to 50% by weight of a copolymer of ethylene and one or more comonomers selected from C._{sub.3}-C._{sub.10} alpha-olefins, containing from 1.0% to 20.0% by weight of alpha-olefin units; and component B._{sup.II}) contains 25.0% by weight or less of a fraction soluble in xylene at 25° C. (XS._{sup.II}), both the amount of alpha-olefin units and of the fraction XS._{sup.II} being referred to the weight of B._{sup.II}); and B._{sup.III}) from 30% to 60% by weight of a copolymer of ethylene and propylene containing from 25.0% to 75.0% by weight of

ethylene derived units; and component B.sup.III) contains from 40.0% to 95.0% by weight of a fraction soluble in xylene at 25° C. (XS.sup.III), both the amount of ethylene units and of the fraction XS.sup.III being referred to the weight of B.sup.III); the amounts of A) and B) being referred to the total weight of A)+B) and the amounts of B.sup.I), B.sup.II) and B.sup.III) being referred to the total weight of B.sup.I)+B.sup.II)+B.sup.III).

(10) In some embodiments, component A) is from 75% to 97% by weight, alternatively from 78% to 95% by weight, alternatively from 85% to 95% by weight, of a copolymer of ethylene with one or more comonomers selected from C.sub.3-C.sub.10 alpha-olefins. In some embodiments, the ethylene copolymer has a density of 0.925 g/cm.sup.3 or higher, alternatively 0.930 g/cm.sup.3 or higher, alternatively 0.925 to 0.945 g/cm.sup.3, alternatively 0.930 to 0.945 g/cm.sup.3, measured according to ISO 1183-1:2012 at 23° C. In some embodiments, the ethylene copolymer has a MI.sub.2 value of 0.5 g/10 min. or greater, alternatively 1 g/10 min. or greater, alternatively 1.5 g/10 min. or greater, where MI.sub.2 is the Melt Index measured according to ISO 1133-2:2011 at 190° C. with a load of 2.16 kg.

(11) In some embodiments, component B) is from 3% to 25% by weight, alternatively from 5% to 22% by weight, alternatively from 5% to 15% by weight, of a polyolefin composition made from or containing components B.sup.I), B.sup.II), and B.sup.III).

(12) In some embodiments, component B.sup.I) is from 5% to 35% by weight of one or more propylene homopolymer(s) or one or more copolymer(s) of propylene with ethylene and/or C.sub.4-C.sub.8 α-olefin(s), containing 90% or more of propylene units, or combinations of the homopolymers and copolymers; and component B.sup.I) contains 10.0% by weight or less, alternatively from 1% to 10% by weight, of a fraction soluble in xylene at 25° C. (XS.sup.I), both the amount of propylene units and of the fraction XS.sup.I being referred to the weight of B.sup.I).

(13) In some embodiments, component B.sup.II) is from 20% to 50% by weight of a copolymer of ethylene and one or more comonomers selected from C.sub.3-C.sub.10 alpha-olefins, containing from 1.0% to 20.0% by weight of alpha-olefin units; and component B.sup.II) contains 25.0% by weight or less, alternatively from 0.5% to 25% by weight, of a fraction soluble in xylene at 25° C. (XS.sup.II), both the amount of alpha-olefin units and of the fraction XS.sup.II being referred to the weight of B.sup.II). In some embodiments, the present disclosure provides a filament or fiber made from or containing the polyethylene composition (I).

(14) In some embodiments, other polyolefin components and/or components different from polyolefins are present in the filament or fiber. In some embodiments, the polyethylene composition (I) constitutes the overall polymer composition present in the filament or fiber, or part of such polymer composition, wherein the total weight of the filament or fiber is the sum of the polyethylene composition (I) and the other components.

Description

DETAILED DESCRIPTION OF THE INVENTION

- (1) As used herein, the term “copolymer” refers to polymers containing one kind or more than one kind of comonomers.
- (2) In some embodiments, the comonomers are alpha-olefins selected from olefins having the formula CH₂=CHR, wherein R is an alkyl radical, linear or branched, or an aryl radical, having the appropriate number of carbon atoms; thus, for instance, from 1 to 8 carbon atoms for C.sub.3-C.sub.10 alpha-olefins, and from 2 to 6 carbon atoms for C.sub.4-C.sub.8 alpha-olefins.
- (3) In some embodiments, the C.sub.3-C.sub.10 alpha-olefins are selected from the group consisting of propylene, butene-1, pentene-1, 4-methylpentene-1, hexene-1, octene-1, and decene-1.
- (4) In some embodiments, alpha-olefin comonomers are present individually or in a mixture with

other alpha-olefin comonomers. In some embodiments, alpha-olefin comonomers are present with ethylene in the copolymer component A). In some embodiments, the alpha-olefin comonomers are C₅-C₁₀ alpha-olefins.

(5) In some embodiments, the comonomers in the copolymer component A) are selected from the group consisting of pentene-1, hexene-1, and octene-1. In some embodiments, the comonomer is hexene-1.

(6) In some embodiments, the upper limit of MI₂ values for the copolymer component A) of the polyethylene composition (I) is 10 g/10 min., alternatively 5 g/10 min.

(7) In some embodiments, the ranges of MI₂ values are: from 0.5 to 10 g/10 min., alternatively from 1 to 10 g/10 min., alternatively from 1.5 to 10 g/10 min., alternatively from 0.5 to 5 g/10 min., alternatively from 1 to 5 g/10 min., alternatively from 1.5 to 5 g/10 min.

(8) In some embodiments, the density values for the copolymer component A) are from 0.925 to 0.938 g/cm³, alternatively from 0.930 to 0.938 g/cm³.

(9) In some embodiments, features for the copolymer component A) are (independently from each other, or in any combination): a Mw/Mn value from 4 to 18, alternatively from 5 to 15, alternatively from 7 to 15, where Mw and Mn are the weight average molecular weight and the number average molecular weight respectively, measured by GPC (Gel Permeation Chromatography); a Mw value of from 20000 g/mol to 500000 g/mol, alternatively from 50000 g/mol to 300000 g/mol, alternatively from 80000 g/mol to 250000 g/mol; a z-average molar mass M_z of less than 1,000,000 g/mol, alternatively from 200000 g/mol to 800000 g/mol, wherein M_z is the z-average molar mass measured by GPC; comonomer content of 8% by weight or lower, alternatively from 8% to 3% by weight, with respect to the total weight of the copolymer; a MI₂₁ value of from 55 to 120 g/10 min., alternatively from 60 to 110 g/10 min., where MI₂₁ is the Melt Index measured according to ISO 1133-2:2011 at 190° C. with a load of 21.6 kg; a MI₂₁/MI₂ value from 25 to 60; a degree of long chain branching λ (lambda) of from 0 to 2 long chain branches/10000 carbon atoms, alternatively from 0.1 to 1.5 long chain branches/10000 carbon atoms; a content of vinyl groups of at least 0.6/1000 carbon atoms, alternatively from 0.6 to 2 vinyl groups/1000 carbon atoms; or a content of vinylidene groups of at least 0.05/1000 carbon atoms, alternatively from 0.1 to 1 vinylidene groups/1000 carbon atoms. In some embodiments, the comonomer content is derived from hexene-1.

(10) In some embodiments, the degree of long chain branching λ (lambda) is measured by light scattering as described in ACS Series 521, 1993, Chromatography of Polymers, Ed. Theodore Provder; Simon Pang and Alfred Rudin: Size-Exclusion Chromatographic Assessment of Long-Chain Branch Frequency in Polyethylenes, page 254-269.

(11) The content of vinyl groups/1000 carbon atoms is determined by means of FT-IR in accordance with ASTM D 6248-98. As used herein, the expression “vinyl groups” refers to —CH=CH₂ groups; vinylidene groups and internal olefinic groups are not encompassed by this expression. It is believed that in some instances, vinyl groups are attributed to a polymer termination reaction after an ethylene insertion. It is further believed that in some instances, vinylidene end groups are formed after a polymer termination reaction after a comonomer insertion.

(12) The determination of the content of vinylidene groups/1000 carbon atoms is carried out in accordance with ASTM D 6248-98.

(13) In some embodiments, the molecular weight distribution of the copolymer component A) is monomodal, bimodal or multimodal. As used herein, the term “monomodal molecular weight distribution” refers to the molecular weight distribution, as determined with Gel Permeation Chromatography (GPC), having a single maximum. In some embodiments, the molecular weight distribution curve of a GPC-multimodal polymer is looked at as the superposition of the molecular weight distribution curves of two or more polymer subfractions or subtypes and shows two or more distinct maxima or is at least distinctly broadened compared with the curves for the individual

fractions. In some embodiments, the molecular weight distribution of the copolymer component A) is monomodal or bimodal. In some embodiments, the molecular weight distribution of the copolymer component A) is monomodal.

(14) In some embodiments, the amount of the copolymer component A) with a molar mass of below 1,000,000 g/mol, as determined by GPC in the standard determination of the molecular weight distribution, is above 95.5% by weight, alternatively above 96% by weight, alternatively above 97% by weight. In some embodiments, the amount of the copolymer component A) with a molar mass of below 1,000,000 g/mol is determined during the molar mass distribution measurement by applying the WIN GPC software of the company 'HS-Entwicklungsgesellschaft für wissenschaftliche Hard-und Software mbH', Ober-Hilbersheim/Germany.

(15) In some embodiments, the copolymer component A) of the polyolefin composition (I) is produced by using a single site catalyst system. In some embodiments, the single site catalyst system is made from or containing a hafnocene catalyst component, alternatively a hafnocene catalyst component and an iron complex having a tridentate ligand, alternatively bearing at least two aryl radicals with each bearing a halogen or tert. alkyl substituent in the ortho-position.

(16) In some embodiments, the single site catalyst systems and the polymerization processes for preparing the copolymer component A) are as described in Patent Cooperation Treaty Publication No. WO2009103516.

(17) In some embodiments, hafnocene catalyst components are cyclopentadienyl complexes. In some embodiments, the cyclopentadienyl complexes are bridged or unbridged biscyclopentadienyl complexes as described in European Patent Publication Nos. EP 129 368, EP 561 479, EP 545 304, and EP 576 970. In some embodiments, the cyclopentadienyl complexes are monocyclopentadienyl complexes. In some embodiments, the monocyclopentadienyl complexes are selected from the group consisting of bridged amidocyclopentadienyl complexes described in European Patent No. EP 416 815, multinuclear cyclopentadienyl complexes described in European Patent No. EP 632 063, pi-ligand-substituted tetrahydropentalenes described in European Patent No. EP 659 758, and pi-ligand-substituted tetrahydroindenes described in European Patent No. EP 661 300.

(18) In some embodiments, hafnocene catalyst components are selected from the group consisting of bis(cyclopentadienyl)hafnium dichloride, bis(indenyl)hafnium dichloride, bis(fluorenyl)hafnium dichloride, bis(tetrahydroindenyl)hafnium dichloride, bis(pentamethylcyclopentadienyl)hafnium dichloride, bis(trimethylsilylcyclopentadienyl)hafnium dichloride, bis(trimethoxy silylcyclopentadienyl)hafnium dichloride, bis(ethylcyclopentadienyl)hafnium dichloride, bis(isobutylcyclopentadienyl)hafnium dichloride, bis(3-butenylcyclopentadienyl)hafnium dichloride, bis(methylcyclopentadienyl)hafnium dichloride, bis(1,3-di-tert-butylcyclopentadienyl)hafnium dichloride, bis(trifluoromethylcyclopentadienyl)hafnium dichloride, bis(tert-butylcyclopentadienyl)hafnium dichloride, bis(n-butylcyclopentadienyl)hafnium dichloride, bis(phenylcyclopentadienyl)hafnium dichloride, bis(N,N-dimethylaminomethylcyclopentadienyl)hafnium dichloride, bis(1,3-dimethylcyclopentadienyl)hafnium dichloride, bis(1-n-butyl-3-methylcyclopentadienyl)hafnium dichloride, (cyclopentadienyl)(methylcyclopentadienyl)hafnium dichloride, (cyclopentadienyl)(n-butylcyclopentadienyl)hafnium dichloride, (methylcyclopentadienyl)(n-butylcyclopentadienyl)hafnium dichloride, (cyclopentadienyl)(1-methyl-3-n-butylcyclopentadienyl)hafnium dichloride, bis(tetramethylcyclopentadienyl)hafnium dichloride, and corresponding dimethylhafnium compounds.

(19) In some embodiments, the hafnocene catalyst component is bis(n-butylcyclopentadienyl)hafnium dichloride.

(20) In some embodiments, iron complexes are selected from the group consisting of 2,6-Bis[1-(4,6-Dimethyl-2-chloro-phenylimino) ethyl]pyridine iron(II) dichloride, 2,6-Bis[1-(2-tert.butylphenylimino)ethyl]pyridine iron(II) dichloride, 2,6-Bis [1-(2-tert.butyl-6-chlorophenylimino)ethyl]pyridine iron(II) dichloride, 2,6-Bis [1-(2-chloro-6-methylphenylimino)ethyl]pyridine iron(II) dichloride, 2,6-Bis [1-(2,4-dichlorophenylimino)ethyl]-

pyridine iron(II) dichloride, 2,6-Bis[1-(2,6-dichlorophenylimino)ethyl]pyridine iron(II) dichloride, 2,6-Bis[1-(2,4-dichlorophenylimino)methyl]pyridine iron(II) dichloride, 2,6-Bis[1-(2,4-dichloro-6-methylphenylimino)ethyl]pyridine iron(II) dichloride, 2,6-Bis[1-(2,4-difluorophenylimino)ethyl]pyridine iron(II) dichloride, 2,6-Bis[1-(2,4-dibromophenylimino)ethyl]pyridine iron(II) dichloride, and respective trichlorides, dibromides or tribromides.

(21) In some embodiments, the molar ratio of the hafnocene compound to the iron complex is in the range from 1:100 to 100:1, alternatively from 1:10 to 10:1, alternatively from 1:1 to 5:1.

(22) In some embodiments, the single site catalyst system is further made from or containing an activating compound. In some embodiments, the activating compound is an aluminoxane. In some embodiments, the aluminoxane is selected from the compounds described in Patent Cooperation Treaty Publication No. WO00/31090.

(23) In some embodiments, the aluminoxane compound is methylaluminoxane.

(24) In some embodiments, the hafnocene compound, the iron complex and the aluminoxane compounds are used in such amounts that the atomic ratio of aluminum from the aluminoxane compounds, including any aluminum alkyl present, to the transition metal from the hafnocene compound, is in the range from 1:1 to 2000:1, alternatively from 10:1 to 500:1, alternatively from 20:1 to 400:1. In some embodiments, the atomic ratio of aluminum from the aluminoxane compounds, including any aluminum alkyl present, to the iron from the iron complex, is in the range from 1:1 to 2000:1, alternatively from 10:1 to 500:1, alternatively from 20:1 to 400:1.

(25) In some embodiments, the catalyst systems are supported on organic or inorganic supports. In some embodiments, the supports are porous polymer particles or silica.

(26) In some embodiments, molecular weight regulators are used. In some embodiments, molecular weight regulators are chain transfer agents. In some embodiments, the chain transfer agents are hydrogen or ZnEt.sub.2.

(27) In some embodiments, the process for producing the copolymer component A) is carried out using polymerization methods at temperatures in the range from -60 to 350° C., alternatively from 0 to 200° C., alternatively from 25 to 150° C., and under pressures of from 0.5 to 4000 bar, alternatively from 1 to 100 bar, alternatively from 3 to 40 bar. In some embodiments, the polymerization is carried out in bulk, in suspension, in the gas phase or in a supercritical medium in reactors used for the polymerization of olefins. In some embodiments, the polymerization is carried out batchwise, alternatively continuously in one or more stages. In some embodiments, the process is selected from the group consisting of solution processes, suspension processes, stirred gas-phase processes and gas-phase fluidized-bed processes.

(28) In some embodiments, the polyolefin composition B) has MIL value from 0.5 to 25 g/10 min., alternatively from 0.8 to 20.0 g/10 min.; alternatively from 1.0 to 18.0g/10 min., where MIL is the Melt Index L, measured according to ISO 1133-2:2011 at 230° C. with a load of 2.16 kg.

(29) In some embodiments, the intrinsic viscosity (I.V.) of the fraction soluble in xylene at 25° C. of the polyolefin composition B), measured in tetrahydronaphthalene at 135° C., is 3.0 dl/g or lower, alternatively 2.8 dl/g or lower, alternatively from 1.5 to 3.0 dl/g, alternatively from 1.5 to 2.8 dl/g.

(30) In some embodiments, component B.sup.I) of the polyolefin composition B) has a MIL value from 50 to 200 g/10 min; alternatively from 80 to 170 g/10 min.

(31) In some embodiments, the MIL value of components B.sup.I)+B.sup.II) blended together is from 0.1 to 70 g/10 min., alternatively from 1 to 50 g/10 min, alternatively from 8 to 40 g/10 min.

(32) In some embodiments, the alpha-olefin comonomers are present individually or in a mixture with other alpha-olefin comonomers. In some embodiments, alpha-olefin comonomers are present with ethylene in component B.sup.II) of the polyolefin composition B). In some embodiments, the alpha-olefin comonomers are selected from the group consisting of propylene and butene-1.

(33) In some embodiments, the alpha-olefin comonomer is butene-1.

- (34) In some embodiments, component B.sup.II) has a density of from 0.940 to 0.965 g/cm.sup.3.
- (35) In some embodiments, the amounts by weight of ethylene derived units in component referred to the weight of are: from 40 to 75%, alternatively from 45 to 75%, alternatively from 48 to 75%, alternatively from 40 to 70%, alternatively from 45 to 70%, alternatively from 48 to 70%.
- (36) In some embodiments and in component B.sup.III), the ratio XS.sup.III/C.sub.2, wherein XS.sup.III is the amount of fraction soluble in xylene and C.sub.2 is the amount by weight of ethylene derived units referred to the weight of is 2.8 or lower, alternatively 2.5 or lower, alternatively 2 or lower, alternatively from 0.8 to 2.8, alternatively 0.8 to 2.5, alternatively 0.8 to 2.
- (37) In some embodiments, the relative amounts of components B.sup.I), B.sup.II) and B.sup.III), in percent by weight with respect to the total weight of B.sup.I)+B.sup.II)+B.sup.III) are: from 10% to 30% of B.sup.I), from 25% to 45% of B.sup.II) and from 35% to 55% of B.sup.III), alternatively from 15% to 23% of B.sup.I), from 30% to 40% of B.sup.II) and from 40% to 50% of B.sup.III).
- (38) In some embodiments, polyolefin composition B) is prepared by a sequential polymerization, including at least three sequential steps, wherein components B.sup.I), B.sup.II) and B.sup.III) are prepared in separate subsequent steps, operating in each step, except the first step, in the presence of the polymer formed and the catalyst used in the preceding step. In some embodiments, the catalyst is added in the first step and not a subsequent step. In some embodiments, the catalyst is active in the first step and the subsequent steps.
- (39) In some embodiments, the polymerization is continuous or batch. In some embodiments, the polymerization is carried out in liquid phase, in the presence or not of inert diluent, or in gas phase, or by mixed liquid-gas techniques. In some embodiments, the polymerization is carried out in gas phase.
- (40) In some embodiments, the polymerization temperature is from 50 to 100° C. In some embodiments, the polymerization pressure is atmospheric or higher.
- (41) In some embodiments, molecular weight is regulated. In some embodiments, the regulation of the molecular weight is carried out by using hydrogen.
- (42) In some embodiments, the polymerizations are carried out in the presence of a Ziegler-Natta catalyst. In some embodiments, a Ziegler-Natta catalyst is made from or containing the product of the reaction of an organometallic compound of group 1, 2 or 13 of the Periodic Table of elements with a transition metal compound of groups 4 to 10 of the Periodic Table of Elements (new notation). In some embodiments, the transition metal compound is selected from the group consisting of compounds of Ti, V, Zr, Cr and Hf. In some embodiments, the transition metal compound is supported on MgCl.sub.2.
- (43) In some embodiments, the catalysts are made from or containing the product of the reaction of the organometallic compound of group 1, 2 or 13 of the Periodic Table of elements, with a solid catalyst component made from or containing a Ti compound and an electron donor compound supported on MgCl.sub.2.
- (44) In some embodiments, the organometallic compounds are aluminum alkyl compounds.
- (45) In some embodiments, the polyolefin composition B) is obtainable by using a Ziegler-Natta polymerization catalyst, alternatively a Ziegler-Natta catalyst supported on MgCl.sub.2, alternatively a Ziegler-Natta catalyst made from or containing the product of reaction of: 1) a solid catalyst component made from or containing a Ti compound and an electron donor (internal electron-donor) supported on MgCl.sub.2; 2) an aluminum alkyl compound (cocatalyst); and, optionally, 3) an electron-donor compound (external electron-donor).
- (46) In some embodiments, the solid catalyst component 1) contains as electron-donor a compound selected from the group consisting of ethers, ketones, lactones, compounds containing N, P and/or S atoms, and mono- and dicarboxylic acid esters.
- (47) In some embodiments, the catalysts are as described in U.S. Pat. No. 4,399,054 and European Patent No. 45977.

(48) In some embodiments, the electron-donor compounds are selected from the group consisting of phthalic acid esters and succinic acid esters. In some embodiments, the phthalic acid ester is diisobutyl phthalate.

(49) In some embodiments, the succinic acid esters are represented by the formula (I):

(50) ##STR00001##

(51) wherein the radicals R.sub.1 and R.sub.2, equal to or different from each other, are a C.sub.1-C.sub.20 linear or branched alkyl, alkenyl, cycloalkyl, aryl, arylalkyl or alkylaryl group, optionally containing heteroatoms; the radicals R.sub.3 to R.sub.6 equal to or different from each other, are hydrogen or a C.sub.1-C.sub.20 linear or branched alkyl, alkenyl, cycloalkyl, aryl, arylalkyl or alkylaryl group, optionally containing heteroatoms. In some embodiments, the radicals R.sub.3 to R.sub.6 are joined to the same carbon atom and linked together to form a cycle.

(52) In some embodiments, R.sub.1 and R.sub.2 are selected from the group consisting of C.sub.1-C.sub.8 alkyl, cycloalkyl, aryl, arylalkyl and alkylaryl groups. In some embodiments, R.sub.1 and R.sub.2 are selected from primary alkyls, alternatively branched primary alkyls. In some embodiments, R.sub.1 and R.sub.2 groups are selected from the group consisting of methyl, ethyl, n-propyl, n-butyl, isobutyl, neopentyl, and 2-ethylhexyl. In some embodiments, R.sub.1 and R.sub.2 groups are selected from the group consisting of ethyl, isobutyl, and neopentyl.

(53) In some embodiments, R.sub.3 to R.sub.5 are hydrogen and R.sub.6 is a branched alkyl, cycloalkyl, aryl, arylalkyl or alkylaryl radical having from 3 to 10 carbon atoms. In some embodiments, at least two radicals from R.sub.3 to R.sub.6 are different from hydrogen and selected from C.sub.1-C.sub.20 linear or branched alkyl, alkenyl, cycloalkyl, aryl, arylalkyl or alkylaryl group, optionally containing heteroatoms. In some embodiments, the two radicals different from hydrogen are linked to the same carbon atom. In some embodiments, R.sub.3 and R.sub.5 are different from hydrogen and linked to different carbon atoms. In some embodiments, R.sub.4 and R.sub.6 are different from hydrogen and linked to different carbon atoms.

(54) In some embodiment, the electron-donors are the 1,3-diethers described in European Patent Application Nos. EP-A-361 493 and 728769.

(55) In some embodiments, cocatalysts 2) are trialkyl aluminum compounds. In some embodiments, the trialkyl aluminum compounds are selected from the group consisting of Al-triethyl, Al-triisobutyl, and Al-tri-n-butyl.

(56) In some embodiments, the electron-donor compounds 3) used as external electron-donors (added to the Al-alkyl compound) are selected from the group consisting of aromatic acid esters, heterocyclic compounds, and silicon compounds containing at least one Si—OR bond (where R is a hydrocarbon radical). In some embodiments, the aromatic acid esters are alkylic benzoates. In some embodiments, the heterocyclic compounds are selected from the group consisting of 2,2,6,6-tetramethylpiperidine and 2,6-diisopropylpiperidine.

(57) In some embodiments, the silicon compounds have the formula

R.sup.1.sub.aR.sup.2.sub.bSi(OR.sup.3)_c, where a and b are integer numbers from 0 to 2, c is an integer from 1 to 3 and the sum (a+b+c) is 4; R.sup.1, R.sup.2 and R.sup.3 are alkyl, cycloalkyl or aryl radicals with 1-18 carbon atoms optionally containing heteroatoms.

(58) In some embodiments, the silicon compounds are selected from the group consisting of (tert-butyl).sub.2Si(OCH.sub.3).sub.2, (cyclohexyl)(methyl)Si(OCH.sub.3).sub.2, (phenyl).sub.2Si(OCH.sub.3).sub.2, and (cyclopentyl).sub.2Si(OCH.sub.3).sub.2.

(59) In some embodiments, the 1,3-diethers are used as external donors. In some embodiments, the internal donor is a 1,3-diethers and the external donor is omitted.

(60) In some embodiments, the catalysts are precontacted with small quantities of olefin (prepolymerization), maintaining the catalyst in suspension in a hydrocarbon solvent, and polymerizing at temperatures from room to 60° C., thereby producing a quantity of polymer from 0.5 to 3 times the weight of the catalyst.

(61) In some embodiments, the operation takes place in liquid monomer, producing a quantity of

polymer up to 1000 times the weight of the catalyst.

(62) In some embodiments, the polyolefin composition B) is prepared as a physical blend of the separately-prepared components rather than as a reactor blend.

(63) In some embodiments, the polyethylene composition (I) is obtainable by melting and mixing the components, and the mixing is effected in a mixing apparatus at temperatures from 180 to 310° C., alternatively from 190 to 280° C., alternatively from 200 to 250° C.

(64) In some embodiments, the melt-mixing apparatuses are extruders or kneaders, alternatively twin-screw extruders. In some embodiments, the components are premixed at room temperature in a mixing apparatus.

(65) In some embodiments, the polyethylene composition (I) was further made from or containing additives. In some embodiments, the additives were selected from the group consisting of stabilizing agents (against heat, light, or U.V.), plasticizers, antiacids, antistatic, water repellant agents, and pigments.

(66) In some embodiments, the filament or fiber is made from or containing at least 70% by weight of the polyethylene composition (I), alternatively at least 80% by weight, alternatively 90% or 95% by weight of polyethylene composition (I), with respect to the total weight of the filament or fiber, the upper limit being 100% in the cases.

(67) In some embodiments, the artificial turf filaments are characterized by a rounded cross-section or an angular cross-section. In some embodiments, the rounded cross-section is selected from the group consisting of circular, oval, lenticular, or multilobal. In some embodiments, the angular cross-section is rectangular.

(68) In some embodiments and as used herein, the filaments having rounded cross-section are “monofilaments”. In some embodiments and as used herein, the filaments having angular cross-section are “tapes”. As used herein, the term “filament” refers to the monofilaments and tapes.

(69) In some embodiments, the tapes have a thickness from 0.03 to 1 mm and width from 2 to 20 mm.

(70) In some embodiments, the filaments have a titer of at least 20 den.

(71) In some embodiments, the titer values for the filaments are at least 50 den, alternatively at least 100 or 200, alternatively at least 500 den. In some embodiments, the upper limit is 3000 den for monofilaments and 25000 den for tapes.

(72) In some embodiments, the filament is stretched by drawing. In some embodiments, the draw ratios are from 1.5 to 10 (1:1.5 to 1:10), alternatively from 3 to 10 (1:3 to 1:10). In some embodiments, the draw ratios apply to the fibers.

(73) In some embodiments, the filaments are made from or containing components made of materials different from polyolefins. In some embodiments, the different materials are embedded reinforcing fibers. In some embodiments, the embedded reinforcing fibers are made from or containing polyamide.

(74) In some embodiments, the filaments are used in the form of bundles for preparation of artificial turf structures. In some embodiments, the number of individual filaments in a single bundle is up to 20. In some embodiments, filaments made of different polymer materials are present in the bundles. In some embodiments, the different polymer materials are polypropylene or polyamide.

(75) In some embodiments, the bundles are held together by one or more wrapping filaments. In some embodiments, the wrapping filaments are made from or containing polymer materials. In some embodiments, the polymer materials are polypropylene or polyethylene. In some embodiments, the wrapping filaments are bonded to one another and/or with the presently-disclosed bundled filaments.

(76) In some embodiments, bundles of filaments are obtained by fibrillation of tapes.

(77) In some embodiments, the process for preparing polyolefin filaments includes the steps of: (a) melting the polyethylene composition (I) and, when present, the other polymer components; (b)

spinning the filaments or extruding a precursor film or tape; (c) optionally drawing the filaments or the precursor film or tape and/or cutting the precursor film or tape and optionally drawing the filaments, when no drawing is previously carried out; and (d) optionally finishing the filaments obtained from step (b) or (c).

(78) In some embodiments, the melting step (a) and the spinning or extrusion step (b) are carried out continuously in sequence by using mono- or twin-screw extruders, equipped with a spinning or extrusion head. In some embodiments, the melt-mixing step is carried out in the same spinning or extrusion apparatus.

(79) In some embodiments, the spinning heads have a plurality of holes with the same shape as the transversal section of the filament (monofilament or tape).

(80) In some embodiments, the film extrusion heads are flat or annular dies used for the film preparation.

(81) In some embodiments, a precursor film or tape is obtained in step (b) and then processed in step (c) by cutting the precursor film or tape into tapes of a certain size. In some embodiments, the drawing treatment is carried out on the precursor film or tape and not on the final filament.

(82) In some embodiments, the finishing treatments are selected from the group consisting of fibrillation and crimping.

(83) In some embodiments, fibrillation is carried out on tapes.

(84) In some embodiments, the melting step (a) and the spinning or extrusion step (b) are carried out at the same temperatures as the melt-mixing step. In some embodiments, the temperatures are from 180 to 310° C., alternatively from 190 to 280° C., alternatively from 200 to 250° C.

(85) In some embodiments, the spinning conditions are: temperature in the extruder head from 200 to 300° C.; take-up speed for primary web (unstretched) from 1 to 50 m/min.

(86) In some embodiments, the film extrusion conditions are: temperature in the extruder head from 200 to 300° C.; output value from 20 to 1000 kg/hour (on industrial plants).

(87) In some embodiments, the filament or the precursor film obtained in step (b) are cooled. In some embodiments, cooling is achieved with one or more chill rolls. In some embodiments, cooling is achieved by immersion in water at a temperature from 5 to 25° C.

(88) To carry out the drawing treatment, the filament (monofilament or tape) or the precursor tape are previously heated at a temperature from 40 to 120-140° C. In some embodiments, heating is achieved by using a hot air oven, a boiling water bath, or heated rolls or by irradiation.

(89) In some embodiments, drawing is achieved by delivering the filament or the precursor tape through a series of rolls having different rotation speeds.

(90) In some embodiments, fibrillation is achieved by feeding the tape between rolls. In some embodiments, the rolls cut longitudinally and/or diagonally.

(91) In some embodiments, fibers with lower denier than filaments are prepared by extruding the polymer melt through the spinning heads, wherein the holes have a smaller diameter with respect to the diameter used for filaments. In some embodiments, the denier of the fibers is under 20 den, alternatively from 1 to 15 den. In some embodiments, the fibers emerging from the spinning head are subsequently subjected to quenching and oriented by stretching.

(92) In some embodiments, the artificial turf is obtained by fixing the filaments or the bundles of filaments to a substrate. As used herein, the term “backing” refers to the substrate.

(93) In some embodiments, the backing is a polyolefin fiber mat. In some embodiments, the polyolefin is polypropylene.

(94) In some embodiments, filling materials are deposited over the backing. In some embodiments, the filling materials are selected from the group consisting of sand and rubber particles.

EXAMPLES

(95) The following analytical methods are used to characterize the polymer compositions.

Density

(96) Determined according to ISO 1183-1:2012 at 23° C.

Melt Index MI.SUB.2., MI.SUB.21 .and MIL

(97) Determined according to ISO 1133-2:2011 with the specified temperature and load.

Intrinsic viscosity I.V.

(98) The sample was dissolved in tetrahydronaphthalene at 135° C. and then poured into a capillary viscometer. The viscometer tube (Ubbelohde type) was surrounded by a cylindrical glass jacket; this setup allowed for temperature control with a circulating thermostatic liquid. The downward passage of the meniscus was timed by a photoelectric device.

(99) The passage of the meniscus in front of the upper lamp started the counter which had a quartz crystal oscillator. The counter stopped as the meniscus passed the lower lamp. The efflux time was registered and converted into a value of intrinsic viscosity through Huggins' equation (Huggins, M. L., J. Am. Chem. Soc., 1942, 64, 2716), using the flow time of the pure solvent at the same experimental conditions (same viscometer and same temperature). A single polymer solution was used to determine I.V.

Molecular Weight Distribution Determination

(100) The determination of the molar mass distributions and the means M_n , M_w , M_z and M_w/M_n derived therefrom was carried out by high-temperature gel permeation chromatography using a method described in ISO 16014-1, -2, -4, issues of 2003. The specifics according to the mentioned ISO standards were as follows: Solvent 1,2,4-trichlorobenzene (TCB), temperature of apparatus and solutions 135° C. and, as concentration detector, a PolymerChar (Valencia, Paterna 46980, Spain) IR-4 infrared detector, for use with TCB. A WATERS Alliance 2000 equipped with a pre-column SHODEX UT-G and separation columns SHODEX UT 806 M (3×) and SHODEX UT 807 (Showa Denko Europe GmbH, Konrad-Zuse-Platz 4, 81829 Muenchen, Germany) connected in series was used.

(101) The solvent was vacuum distilled under Nitrogen and stabilized with 0.025% by weight of 2,6-di-tert-butyl-4-methylphenol. The flowrate used was 1 ml/min. The injection was 500µl. The polymer concentration was in the range of 0.01%<conc.<0.05% w/w. The molecular weight calibration was established by using monodisperse polystyrene (PS) standards from Polymer Laboratories (now Agilent Technologies, Herrenberger Str. 130, 71034 Boeblingen, Germany)) in the range from 580 g/mol up to 11600000 g/mol and additionally with Hexadecane.

(102) The calibration curve was then adapted to Polyethylene (PE) by the Universal Calibration method (Benoit H., Rempp P. and Grubisic Z., & in J. Polymer Sci., Phys. Ed., 5, 753(1967)). The Mark-Houwink parameters used were for PS: $k_{sub}PS=0.000121$ dl/g, $\alpha_{sub}PS=0.706$ and for PE $k_{sub}PE=0.000406$ dl/g, $\alpha_{sub}PE=0.725$, valid in TCB at 135° C. Data recording, calibration and calculation were carried out using NTGPC_Control_V6.02.03 and NTGPC_V6.4.24 (hs GmbH, HauptstraBe 36, D-55437 Ober-Hilbersheim, Germany) respectively.

Comonomer Content

(103) The comonomer content was determined by IR in accordance with ASTM D 6248 98, using an FT-IR spectrometer Tensor 27 from Bruker, calibrated with a chemometric model for determining ethyl- or butyl-side-chains in PE for butene or hexene as comonomer, respectively. The result was compared to the estimated comonomer content derived from the mass-balance of the polymerization process and found to agree.

Tenacity and Elongation at Break of Filaments

(104) Each filament was fixed to the clamps of an Instron dynamometer (model 1122) and tensioned to break with a traction speed of 20 mm/min for elongations lower than 100% and 50 mm/min for elongations greater than 100%, the initial distance between the clamps being of 20 mm. The ultimate strength (load at break) and the elongation at break were determined.

(105) The tenacity was derived using the following equation:

Tenacity=Ultimate strength (cN)×10/Titer (den).

Elastic Recovery of Filaments

(106) Elastic recovery was measured with a dynamometer at 23° C.

(107) Two deformations cycles were applied to the samples under the following conditions:

First Cycle

(108) Initial Cross head distance 200 mm, cross head speed 3.3 mm/s, maximum cross head distance at maximum deformation 400 mm relaxation time at maximum cross head distance 60 seconds reverse cross head speed 3.3 mm/second cross head time at the end of the deformation cycle (cross head distance 200 mm) 180 seconds.

Second Cycle—Carried Out Three Minutes after the End of the First Cycle

(109) Initial Cross head distance 200 mm, cross head speed 3.3 mm/s.

Third to Fifth Cycles

(110) Carried out under the same conditions as for the second cycle, with three minutes intervals after the second and subsequent cycles.

(111) Percent of residual deformation was the percentage value of the deformation recorded when the load cell started to record a tensional force in the second cycle corresponding to the formula
Percent of residual deformation = $100 \times (\text{current crosshead distance} - \text{initial crosshead distance}) / (\text{initial crosshead distance})$. end of test

(112) Elastic recovery (%) was the percentage value of the maximum deformation (400 mm, corresponding to 100% deformation with respect to the initial cross head distance of 200 mm) minus the percent of residual deformation.

Xylene-Soluble Fraction at 25° C. (XS)

(113) The Xylene Soluble fraction was measured according to ISO 16152, 2005, but with the following deviations (the parentheses provides that which was prescribed by the ISO 16152)

(114) The solution volume was 250 ml (200 ml);

(115) During the precipitation stage at 25° C. for 30 min, the solution, for the final 10 minutes, was kept under agitation by a magnetic stirrer (30 min, without any stirring at all);

(116) The final drying step was done under vacuum at 70° C. (100° C.);

(117) The content of the xylene-soluble fraction was expressed as a percentage of the original 2.5 grams and then, by difference (complementary to 100), the xylene insoluble %;

(118) XS of components B.sup.II) and B.sup.III) were calculated by using the formula;

(119) $XS_{\text{sub.tot}} = W_{\text{sub.B.sup.I}} + W_{\text{sub.B.sup.II}} + W_{\text{sub.B.sup.III}}$

(120) wherein $W_{\text{sub.B.sup.I}}$, $W_{\text{sub.B.sup.II}}$ and $W_{\text{sub.B.sup.III}}$ were the relative amount of components B.sup.I), B.sup.II) and B.sup.III) ($B_{\text{sup.I}} + B_{\text{sup.II}} + B_{\text{sup.III}} = 1$)

Examples 1 and 2

(121) The following materials were used to prepare the polyethylene composition (I).

(122) Copolymer Component A)

(123) Ethylene copolymer was prepared as described in Example 4 of Patent Cooperation Treaty Publication No. WO2009103516, having the properties reported in Table I below.

(124) TABLE-US-00001 TABLE I Density [g/cm³] 0.936 Mw [g/mol] 96000 Mw/Mn 9 Mz [g/mol] 281000 GPC % at molar mass 1 Mio — C_{sub.6-} content [% by weight] 5.5 MI_{sub.2} [g/10 min.] 2.5 MI_{sub.21} [g/10 min.] 85 Note: C_{sub.6-} = hexene-1

Polyolefin Composition B)

(125) Polyolefin composition was prepared by sequential polymerization.

Solid Catalyst Component

(126) The solid catalyst component used in polymerization was a Ziegler-Natta catalyst component supported on magnesium chloride, containing titanium and diisobutylphthalate as internal donor. An initial amount of microspheroidal MgCl_{sub.2}.Math.2.8C_{sub.2}H_{sub.5}OH was prepared according to the method described in Example 2 of U.S. Pat. No. 4,399,054 but operating at 3,000 rpm instead of 10,000. The resulting adduct was then subjected to thermal dealcoholation at increasing temperatures from 30 to 130° C. in a nitrogen current until the molar alcohol content per mol of Mg was 1.16. Into a 1000 mL four-necked round flask, purged with nitrogen, 500 mL of TiCl_{sub.4} were introduced at 0° C. While stirring, 30 grams of the microspheroidal

MgCl.sub.2.Math.1.16C.sub.2H.sub.5OH adduct were added. The temperature was raised to 120° C. and maintained for 60 minutes. During the temperature increase, an amount of diisobutylphthalate was added to produce a Mg/diisobutylphthalate molar ratio of 18. After the 60 minutes, the stirring was stopped, the liquid was siphoned off, and the treatment with TiCl.sub.4 was repeated at 100° C. for 1 hour in the presence of an amount of diisobutylphthalate to produce a Mg/diisobutylphthalate molar ratio of 27. The stirring was stopped. The liquid was siphoned off, and the treatment with TiCl.sub.4 was repeated at 100° C. for 30 min. After sedimentation and siphoning at 85° C., the solid was washed six times with anhydrous hexane (6×100 ml) at 60° C.

Catalyst System and Prepolymerization

(127) Before introducing the solid catalyst component into the polymerization reactors, the solid catalyst component was contacted at 30° C. for 9 minutes with aluminum triethyl (TEAL) and dicyclopentyldimethoxysilane (DCPMS), in a TEAL/DCPMS weight ratio equal to about 15 and in such quantity that the TEAL/solid catalyst component weight ratio was equal to 4.

(128) The catalyst system was then subjected to prepolymerization by suspending the catalyst system in liquid propylene at 50° C. for about 75 minutes before introducing the catalyst system into the first polymerization reactor.

Polymerization

(129) The polymerization was carried out continuously in a series of three gas-phase reactors equipped with devices to transfer the product from the first reactor to the subsequent reactors. Into the first gas phase polymerization reactor, a propylene-based polymer, corresponding to component B.sup.I), was produced by feeding, in a continuous and constant flow, the prepolymerized catalyst system, hydrogen (used as molecular weight regulator) and propylene, with the components in a gas state. The propylene-based polymer coming from the first reactor was discharged in a continuous flow and, after having been purged of unreacted monomers, introduced, in a continuous flow, into the second gas phase reactor, together with quantitatively constant flows of hydrogen and ethylene, with the components in a gas state. In the second reactor, a copolymer of ethylene, corresponding to component B.sup.II), was produced. The product coming from the second reactor was discharged in a continuous flow and, after having been purged of unreacted monomers, introduced, in a continuous flow, into the third gas phase reactor, together with quantitatively constant flows of hydrogen, ethylene and propylene, with the components in a gas state. In the third reactor, an ethylene-propylene polymer, corresponding to component B.sup.III), was produced. Polymerization conditions, molar ratio of the reactants and composition and properties of the copolymers obtained are shown in Table II. The polymer particles exiting the third reactor were subjected to a steam treatment to remove the reactive monomers and volatile substances, and then dried. Thereafter the polymer particles were mixed with a stabilizing additive composition in a twin screw extruder Berstorff ZE 25 (length/diameter ratio of screws: 34) and extruded under nitrogen atmosphere in the following conditions: Rotation speed: 250 rpm; Extruder output: 15 kg/hour; Melt temperature: 245° C.

(130) The stabilizing additive composition was made of the following components: 0.1% by weight of Irganox® 1010; 0.1% by weight of Irgafos® 168; and 0.04% by weight of DHT-4A (hydrotalcite);

(131) where the percent amounts refer to the total weight of the polymer and stabilizing additive composition.

(132) The Irganox® 1010 was 2,2-bis [3-[5-bis(1,1-dimethylethyl)-4-hydroxyphenyl]-1-oxopropoxy]methyl-1,3-propanediyl-3,5-bis(1,1-dimethylethyl)-4-hydroxybenzene-propanoate, while Irgafos® 168 was tris(2,4-di-tert.-butylphenyl)phosphite. The characteristics relating to the polymer composition are reported in Table II.

(133) TABLE-US-00002 TABLE II Example 1 1.sup.st Reactor - component B.sup.I) Temperature ° C. 60 Pressure barg 16 H.sub.2/C.sub.3— mol. 0.24 Split wt % 22 Xylene soluble of B.sup.I) (XS.sup.I) wt % 4.2 MIL of B.sup.I) g/10 min. 110 2.sup.nd Reactor - component B.sup.II)

Temperature ° C. 80 Pressure barg 18 H.sub.2/C.sub.2— mol. 0.81 C.sub.4—/(C.sub.2— + C.sub.4—) mol. 0.25 C.sub.2—/(C.sub.2— + C.sub.3—) mol. 0.98 Split wt % 32 C.sub.2— content of B.sup.II) * wt % 90 C.sub.4— content of B.sup.II) * wt % 10 Xylene soluble of B.sup.II) (XS.sup.II) * wt % 16.0 Xylene soluble of B.sup.I) + B.sup.II) wt % 12.0 MIL of B.sup.I) + B.sup.II) g/10 min. 35.9 3.sup.rd Reactor - component B.sup.III) Temperature ° C. 65 Pressure barg 18 H.sub.2/C.sub.2— mol. 0.17 C.sub.2—/(C.sub.2— + C.sub.3—) mol. 0.42 Split wt % 46 C.sub.2— content of B.sup.III) * wt % 52 Xylene soluble of B.sup.III) (XS.sup.III) * wt % 83 Total Composition - Component B) MIL g/10 min 1.61 IV on soluble in Xylene dl/g 2.4 Note; C.sub.2— = ethylene; C.sub.3— = propylene; C.sub.4— = 1-butene (IR); split = amount of polymer produced in the concerned reactor. * Calculated values

(134) Component A) and the polyolefin composition B) were blended together using the same extrusion apparatus and conditions for the polyolefin composition B), thereby obtaining the polyethylene composition (I).

(135) The resulting polyethylene composition (I) was spun into filaments with lenticular cross-section.

(136) The apparatus used was an extruder Leonard, 25 mm diameter, 27 L/D long+Gear pump. The die had 8 holes, lenticular shaped, approx. 8.0 mm×1.2 mm.

(137) The main process conditions were: Temperature profile: Cylinder 180-185-190-195° C.; Pump 200° C. Adapter 205° C. Head-die 210° C. Melt temperature: 212+/-3° C.; Output used: around 4 kg/h; Cooling water bath: 21+/-1° C.; Stretching oven set: 106+/-2° C. (hot air); Stretching ratio used: 1:4; Annealing oven set: 106+/-2° C. (hot air); Annealing factor: average -5.0% (slower).

Comparison Examples 1-3

(138) Carried out as in Example 1, except that in Comparison Example 1, component A) was subjected to spinning and testing, in the absence of component B), while, in Comparison Example 2, component B) was replaced with a polyolefin composition C) made from or containing: 31% by weight of a crystalline propylene/ethylene copolymer, having an ethylene content of 3.3% by weight and solubility in xylene at room temperature of 5.5% by weight, and 69% by weight of a propylene/ethylene elastomeric copolymer, having an ethylene content of 27% by weight, solubility in xylene at room temperature of 90.3% by weight.

(139) The polyolefin composition was obtained as a reactor grade, in subsequent stages carried out in gas phase, using a Ziegler-Natta catalyst and had an I.V. of the fraction soluble in xylene at 25° C. of 3.20 dl/g.

(140) The properties of the resulting filaments are reported in Table III.

(141) TABLE-US-00003 TABLE III Example No. Com. Comp. Comp. 1 2 1 2 3 Amount of A) 90 80 100 90 80 [% by weight] Amount of B) 10 20 0 0 0 [% by weight] Amount of C) 0 0 0 10 20 [% by weight] Titer [den.] 2050 2055 2030 2060 1925 Tenacity [g/den] 1.48 1.44 1.47 1.45 2.56 Elongation at Break [%] 155 140 155 145 105 Elastic Recovery [%] 68.1 62.4 61.2 61.9 45

Claims

1. A polyethylene composition comprising: A) from 75% to 97% by weight of a copolymer of ethylene with one or more comonomers selected from C.sub.3-C.sub.10 alpha-olefins, wherein the copolymer has: 1) a density of 0.925 g/cm.sup.3 or higher, measured according to ISO 1183-1:2012 at 23° C.; and 2) a MI.sub.2 value of 0.5 g/10 min. or greater, where MI.sub.2 is the Melt Index measured according to ISO 1133-2:2011 at 190° C. with a load of 2.16 kg; B) from 3% to 25% by weight of a polyolefin composition comprising: B.sup.I) from 5% to 35% by weight of one or more propylene homopolymer(s) or one or more copolymer(s) of propylene with ethylene and/or C.sub.4-C.sub.8 alpha-olefin(s), containing 90% or more of propylene units, or combinations of the homopolymers and copolymers; and component B.sup.I) contains 10.0% by weight or less of a

fraction soluble in xylene at 25° C. (XS.sup.I), both the amount of propylene units and of the fraction XS.sup.I being referred to the weight of B.sup.I); B.sup.II) from 20% to 50% by weight of a copolymer of ethylene and a one or more comonomers selected from C.sub.3-C.sub.10 alpha-olefins, containing from 1.0% to 20.0% by weight of alpha-olefin units; and component B.sup.II) contains 25.0% by weight or less of a fraction soluble in xylene at 25° C. (XS.sup.II), both the amount of alpha-olefin units and of the fraction XS.sup.II being referred to the weight of B.sup.II); and B.sup.III) from 30% to 60% by weight of a copolymer of ethylene and propylene containing from 25.0% to 75.0% by weight of ethylene derived units; and component B.sup.III) contains from 40.0% to 95.0% by weight of a fraction soluble in xylene at 25° C. (XS.sup.III), both the amount of ethylene units and of the fraction XS.sup.III being referred to the weight of B.sup.III); the amounts of A) and B) being referred to the total weight of A)+B) and the amounts of B.sup.I), B.sup.II) and B.sup.III) being referred to the total weight of B.sup.I)+B.sup.II)+B.sup.III).

2. The polyethylene composition of claim 1, wherein component A) is a copolymer of ethylene with hexene-1.

3. The polyethylene composition of claim 1, wherein component A) has MI.sub.2 value from 0.5 to 5 g/10 min.

4. The polyethylene composition of claim 1, wherein component A) has a MI.sub.21/MI.sub.2 value from 25 to 60.

5. The polyethylene composition of claim 1, wherein component A) has a -Mw/Mn value from 4 to 18, where Mw and Mn are the weight average molecular weight and the number average molecular weight respectively, measured by GPC.

6. The polyethylene composition of claim 1, wherein component A) has a z-average molar mass Mz of less than 1000000 g/mol, measured by GPC.

7. The polyethylene composition of claim 1, wherein the intrinsic viscosity (I.V.) of the fraction soluble in xylene at 25° C. of the polyolefin composition B), measured in tetrahydronaphthalene at 135° C., is 3.0 dl/g or lower.

8. The polyethylene composition of claim 1, wherein component B.sup.I) has a MIL value from 50 to 200 g/10 min.

9. The polyethylene composition of claim 1, wherein component B.sup.II) is a copolymer of ethylene with butene-1.

10. The polyethylene composition of claim 1, wherein, in component B.sup.III) the ratio XS.sup.III/C.sub.2, wherein C.sub.2 is the amount by weight of ethylene units referred to the weight of B.sup.III), is 2.8 or lower.

11. An article of manufacture comprising the polyethylene composition of claim 1, wherein the article of manufacture is selected from the group consisting of filaments and fibers.

12. The article of manufacture of claim 11, wherein the filament or fiber is stretched by drawing with a draw ratio from 1.5 to 10.

13. The article of manufacture of claim 11, wherein the article of manufacture is a filament having a titer of at least 20 den.

14. An artificial turf comprising the article of manufacture of claim 11, wherein the article of manufacture is a filament.
